Chicago Quant Alley Assignment 3 : Bandit Algorithms

Objective

The objective of this assignment is to implement and analyze various algorithms in the **multi-armed bandit** (MAB) setting. These algorithms represent a variety of exploration vs. exploitation strategies, and include both classical and modern algorithms for stochastic, adversarial, and contextual environments.

Part 1: Algorithm Implementation

Implement the following algorithms from scratch in Python. Do **not** use libraries such as `scikit-learn` or `mab` for the core logic.

Basic and Classical Bandits

1. **Epsilon-Greedy**
   A simple stochastic algorithm. With probability ϵ, explore randomly; with probability $1 - \epsilon$, exploit the arm with the highest estimated reward.
   [CS340 Lecture Notes].

2. **Explore-Then-Commit (ETC)**
   Pull each arm uniformly for a fixed number of times (exploration phase), then switch to the best arm for the rest of the horizon (commit phase).
   [Even-Dar et al., 2006].

3. **UCB1 (Upper Confidence Bound)**
   Auer et al., 2002 — Finite-Time Analysis of the Multiarmed Bandit Problem.

4. **KL-UCB**
   Garivier and Cappé, 2011 — The KL-UCB Algorithm for Stochastic Bandits.

5. **Thompson Sampling**
   Agrawal and Goyal, 2012 — Analysis of Thompson Sampling for the Multi-armed Bandit Problem.

Adversarial Bandits and Expert Advice

6. **Weighted Majority (WM)**
   Littlestone and Warmuth, 1994 — The Weighted Majority Algorithm.

7. **Exp3 (Exponential-weight Algorithm)**
   Auer et al., 2002 — The Nonstochastic Multiarmed Bandit Problem.

Contextual Bandits

8. **LinUCB (Linear UCB)**
   Li et al., 2010 — A Contextual-Bandit Approach to Personalized News Article Recommendation.

Pure Exploration / Best Arm Identification

9. **Halving Algorithm**
   Karnin et al., 2013 — Almost Optimal Exploration in Multi-Armed Bandits.

10. **LUCB (Lower Upper Confidence Bound)**
    Kalyanakrishnan et al., 2012 — PAC Subset Selection in Stochastic Multi-armed Bandits.

11. **KL-LUCB**
    Kaufmann and Kalyanakrishnan, 2013 — Information Complexity in Bandit Subset Selection.

12. **lil'UCB**
    Jamieson et al., 2014 — lil'UCB: An Optimal Exploration Algorithm for Multi-Armed Bandits.

Part 2: Experimental Evaluation

Conduct experiments in synthetic settings:

- Number of arms: $K = 10$
- Reward distributions: Bernoulli or Gaussian
- Horizon: $T = 10^4$ steps

For contextual bandits:

- Generate feature vectors of fixed dimension
- Use linearly separable reward models

Evaluate:

- Cumulative regret over time
- Frequency of best arm selection
- Confidence bounds vs. observed rewards

Submission Instructions

- Python code (Jupyter notebooks or scripts)
- A PDF report containing:
  - Brief description of each algorithm
  - Implementation approach
  - Experimental plots and analysis
  - Final comparison and summary table
- README file with usage instructions